DBMS WEEK 6 PROJECT REPORT

Product Review and Rating Management System

1. Introduction

The Product Review and Rating Management System is designed for an e-commerce company to collect customer feedback about purchased products. Customers can provide ratings and reviews after completing their orders. The system stores customer opinions, analyses product performance, and identifies highly rated products.

This project uses a handmade resin art store as an example, selling resin art keychains, coasters, pendants, photo frames, bookmarks, name plates, earrings, and decorative trays.

Objectives

- To create a review and rating table.
- To store customer feedback and ratings from 1 to 5 stars.
- To update and delete reviews when required.
- To display customer names along with their reviews.
- To calculate average product ratings.
- To identify highly rated products and products requiring improvement.

Create the REVIEW Table

```
SQL> CREATE TABLE REVIEW (
  2      REVIEWID VARCHAR2(7) PRIMARY KEY,
  3      CUSTOMERID NUMBER(38),
  4      PRODUCTID NUMBER,
  5      RATING NUMBER(1),
  6      REVIEWTEXT VARCHAR2(50),
  7      REVIEWDATE DATE,
  8      CONSTRAINT FK_REVIEW_CUSTOMER
  9          FOREIGN KEY (CUSTOMERID)
 10          REFERENCES CUSTOMER(CUSTOMER_ID),
 11      CONSTRAINT FK_REVIEW_PRODUCT
 12          FOREIGN KEY (PRODUCTID)
 13          REFERENCES PRODUCT(PRODUCT_ID),
 14      CONSTRAINT CHECK_RATING
 15          CHECK (RATING BETWEEN 1 AND 5)
 16  );

Table created.

SQL> DESC REVIEW;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
--------
 REVIEWID                                  NOT NULL VARCHAR2(7)
 CUSTOMERID                                         NUMBER(38)
 PRODUCTID                                          NUMBER
 RATING                                             NUMBER(1)
 REVIEWTEXT                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

```
SQL> SELECT CUSTOMER_ID, C_NAME
  2  FROM CUSTOMER
  3  ORDER BY CUSTOMER_ID;

CUSTOMER_ID C_NAME
----------- --------------------
        101 Arun
        102 Priya
        103 Kumar
        104 Divya
        105 Karthik
        201 DEVI
        202 DIVI
        203 KAVYA

8 rows selected.
```

INSERT CUSTOMER REVIEWS

```
SQL> INSERT INTO REVIEW
  2  VALUES ('R000004', 201, 206, 5,
  3          'Excellent resin art quality', SYSDATE);

1 row created.

SQL> INSERT INTO REVIEW
  2  VALUES ('R000006', 103, 206, 4, 'Good handmade keychain', SYSDATE);

1 row created.

SQL> INSERT INTO REVIEW
  2  VALUES ('R000007', 104, 207, 3, 'Average coaster quality', SYSDATE);

1 row created.

SQL> INSERT INTO REVIEW
  2  VALUES ('R000008', 105, 208, 5, 'Excellent pendant design', SYSDATE);

1 row created.

SQL> INSERT INTO REVIEW
  2  VALUES ('R000009', 201, 209, 4, 'Attractive photo frame', SYSDATE);

1 row created.

SQL> INSERT INTO REVIEW
  2  VALUES ('R000010', 202, 210, 5, 'Amazing resin bookmark', SYSDATE);

1 row created.

SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT REVIEWID, REVIEWTEXT, REVIEWDATE
  2  FROM REVIEW
  3  ORDER BY REVIEWID;

REVIEWI REVIEWTEXT                                          REVIEWDAT
------- ------------------------------------------------- ---------
R000004 Excellent resin art quality                         15-SEP-26
R000006 Good handmade keychain                              16-SEP-26
R000007 Average coaster quality                             17-SEP-26
R000008 Excellent pendant design                            18-SEP-26
R000009 Attractive photo frame                              19-SEP-26
R000010 Amazing resin bookmark                              20-OCT-26

6 rows selected.

SQL> COMMIT;
```

```
SQL> SELECT * FROM REVIEW;

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                          REVIEWDAT
-------------------------------------------------- ---------
R000004        201        206          5
Excellent resin art quality                         15-SEP-26

R000006        103        206          4
Good handmade keychain                              16-SEP-26

R000007        104        207          3
Average coaster quality                             17-SEP-26


REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                          REVIEWDAT
-------------------------------------------------- ---------
R000008        105        208          5
Excellent pendant design                            18-SEP-26

R000009        201        209          4
Attractive photo frame                              19-SEP-26

R000010        202        210          5
Amazing resin bookmark                              20-OCT-26


6 rows selected.
```

UPDATE A REVIEW COMMENT

```
SQL> UPDATE REVIEW
  2  SET REVIEWTEXT = 'Excellent resin art quality'
  3  WHERE REVIEWID = 'R000004';

1 row updated.
```

DELETE A REVIEW

```
1 row updated.

SQL> DELETE FROM REVIEW
  2  WHERE REVIEWID = 'R000004';

1 row deleted.
```

Display all reviews for a product

```
SQL> SELECT *
  2  FROM REVIEW
  3  WHERE PRODUCTID = 206;

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                          REVIEWDAT
--------------------------------------------------- ---------
R000006        103        206          4
Good handmade keychain                              16-SEP-26
```

```
SQL> SELECT C.C_NAME, R.REVIEWTEXT, R.RATING
  2  FROM CUSTOMER C
  3  JOIN REVIEW R
  4  ON C.CUSTOMER_ID = R.CUSTOMERID;

C_NAME               REVIEWTEXT
-------------------- --------------------------------
    RATING
----------
DEVI                 Attractive photo frame
         4

DIVI                 Amazing resin bookmark
         5

Kumar                Good handmade keychain
         4


C_NAME               REVIEWTEXT
-------------------- --------------------------------
    RATING
----------
Divya                Average coaster quality
         3

Karthik              Excellent pendant design
         5
```

```
SQL> SELECT * FROM REVIEW WHERE PRODUCTID = '209';

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
R000009        201        209          4
Attractive photo frame                             19-SEP-26
```

```
SQL> SELECT C.C_NAME,
  2         R.REVIEWTEXT,
  3         R.RATING
  4  FROM CUSTOMER C
  5  JOIN REVIEW R
  6  ON C.CUSTOMER_ID = R.CUSTOMERID;

C_NAME               REVIEWTEXT
-------------------- ------------------------------------
    RATING
----------
DEVI                 Attractive photo frame
         4

DIVI                 Amazing resin bookmark
         5

Kumar                Good handmade keychain
         4


C_NAME               REVIEWTEXT
-------------------- ------------------------------------
    RATING
----------
Divya                Average coaster quality
         3

Karthik              Excellent pendant design
         5
```

Find products having the maximum number of reviews

```
SQL> SELECT P.PRODUCT_NAME,
  2         COUNT(R.REVIEWID) AS TOTAL_REVIEWS
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVIEWS DESC;

PRODUCT_NAME
--------------------------------------------------
TOTAL_REVIEWS
-------------
Resin Art Bookmark
            1

Resin Art Photo Frame
            1

Resin Art Coaster Set
            1


PRODUCT_NAME
--------------------------------------------------
TOTAL_REVIEWS
-------------
Resin Art Pendant
            1

Resin Art Keychain
            1
```

```
SQL> SELECT REVIEWID,
  2         CUSTOMERID,
  3         PRODUCTID,
  4         RATING,
  5         REVIEWTEXT,
  6         REVIEWDATE
  7  FROM REVIEW
  8  ORDER BY REVIEWDATE DESC;

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
R000010        202        210          5
Amazing resin bookmark                             20-OCT-26

R000009        201        209          4
Attractive photo frame                             19-SEP-26

R000008        105        208          5
Excellent pendant design                           18-SEP-26


REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
R000007        104        207          3
Average coaster quality                            17-SEP-26

R000006        103        206          4
Good handmade keychain                             16-SEP-26
```

```
SQL> SELECT *
  2  FROM REVIEW
  3  WHERE RATING > 4;

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
R000008        105        208          5
Excellent pendant design                           18-SEP-26

R000010        202        210          5
Amazing resin bookmark                             20-OCT-26
```

```
SQL> SELECT P.PRODUCT_NAME,
  2         COUNT(R.REVIEWID) AS TOTAL_REVIEWS
  3  FROM PRODUCT P
  4  LEFT JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------
--------
TOTAL_REVIEWS
-------------
Resin Art Bookmark
            1

Resin Art Photo Frame
            1

 RESIN KEYCHAIN
            0


PRODUCT_NAME
--------------------------------------------------
--------
TOTAL_REVIEWS
-------------
Resin Art Keychain
            1

Resin Art Pendant
            1

Resin Pendant
```

```
SQL> SELECT PRODUCTID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PRODUCTID;

 PRODUCTID AVERAGE_RATING
---------- --------------
       207              3
       209              4
       210              5
       206              4
       208              5
```

```
Resin Pendant
            0


PRODUCT_NAME
--------------------------------------------------------------------------------
--------
TOTAL_REVIEWS
-------------
Resin Art Coaster Set
            1

RESIN BOOKMARK
            0


8 rows selected.
```

```
SQL> SELECT P.PRODUCT_NAME,
  2         AVG(R.RATING) AS AVERAGE_RATING
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME
--------------------------------------------------
--------
AVERAGE_RATING
--------------
Resin Art Pendant
             5

Resin Art Bookmark
             5

Resin Art Photo Frame
             4


PRODUCT_NAME
--------------------------------------------------
--------
AVERAGE_RATING
--------------
Resin Art Keychain
             4

Resin Art Coaster Set
             3
```

Identify products with an average rating above 4

```
SQL> SELECT P.PRODUCT_NAME,
  2         AVG(R.RATING) AS AVERAGE_RATING
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME
  7  HAVING AVG(R.RATING) > 4;

PRODUCT_NAME
--------------------------------------------------
--------
AVERAGE_RATING
--------------
Resin Art Bookmark
             5

Resin Art Pendant
             5
```

Report 1 — Product Rating Analysis

Purpose: Display the product name, number of reviews, and average rating.

```
SQL> SELECT P.PRODUCT_NAME,
  2         COUNT(R.REVIEWID) AS TOTAL_REVIEWS,
  3         NVL(AVG(R.RATING), 0) AS AVERAGE_RATING
  4  FROM PRODUCT P
  5  LEFT JOIN REVIEW R
  6  ON P.PRODUCT_ID = R.PRODUCTID
  7  GROUP BY P.PRODUCT_NAME
  8  ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME
--------------------------------------------------------------------
--------
TOTAL_REVIEWS AVERAGE_RATING
------------- --------------
Resin Art Pendant
            1              5

Resin Art Bookmark
            1              5

Resin Art Photo Frame
            1              4
```

Report 2 — Customer Feedback Analysis

This report identifies the most-reviewed products, highly rated products, and products requiring improvement.
Products requiring improvement — average rating below 3

```
PRODUCT_NAME
--------------------------------------------------
--------
TOTAL_REVIEWS AVERAGE_RATING
------------- --------------
Resin Art Keychain
            1              4

Resin Art Coaster Set
            1              3

 RESIN KEYCHAIN
            0              0


PRODUCT_NAME
--------------------------------------------------
--------
TOTAL_REVIEWS AVERAGE_RATING
------------- --------------
Resin Pendant
            0              0

RESIN BOOKMARK
            0              0


8 rows selected.
```

```
SQL> SELECT P.PRODUCT_NAME,
  2         COUNT(R.REVIEWID) AS TOTAL_REVIEWS
  3  FROM PRODUCT P
  4  LEFT JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVIEWS DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
--------
TOTAL_REVIEWS
-------------
Resin Art Bookmark
            1

Resin Art Photo Frame
            1

Resin Art Keychain
            1
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
--------
TOTAL_REVIEWS
-------------
Resin Art Coaster Set
            1

Resin Art Pendant
            1

Resin Pendant
            0


PRODUCT_NAME
--------------------------------------------------------------------------------
--------
TOTAL_REVIEWS
-------------
RESIN BOOKMARK
            0

 RESIN KEYCHAIN
            0


8 rows selected.
```

Report 3 — Rating Distribution

Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 5;

FIVE_STAR_RATINGS
-----------------
                3
```

Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 4;

FOUR_STAR_RATINGS
-----------------
                2
```

```
SQL> SELECT COUNT(*) AS LOW_RATINGS
  2  FROM REVIEW
  3  WHERE RATING <= 2;

LOW_RATINGS
-----------
          0
```

Number of low ratings (1 or 2 stars)

Complete rating distribution

```
SQL> SELECT RATING,
  2         COUNT(*) AS TOTAL_REVIEWS
  3  FROM REVIEW
  4  GROUP BY RATING
  5  ORDER BY RATING DESC;

    RATING TOTAL_REVIEWS
---------- -------------
         5             3
         4             2
         3             1
```

Conclusion

The Product Review and Rating Management System provides a structured method for collecting and analysing customer feedback. By connecting the CUSTOMER, PRODUCT, and REVIEW tables, the company can maintain accurate review records and calculate product ratings.